Introduction to Morphological Operators

EROSION

topics

Introduction to **Morphological Operators**

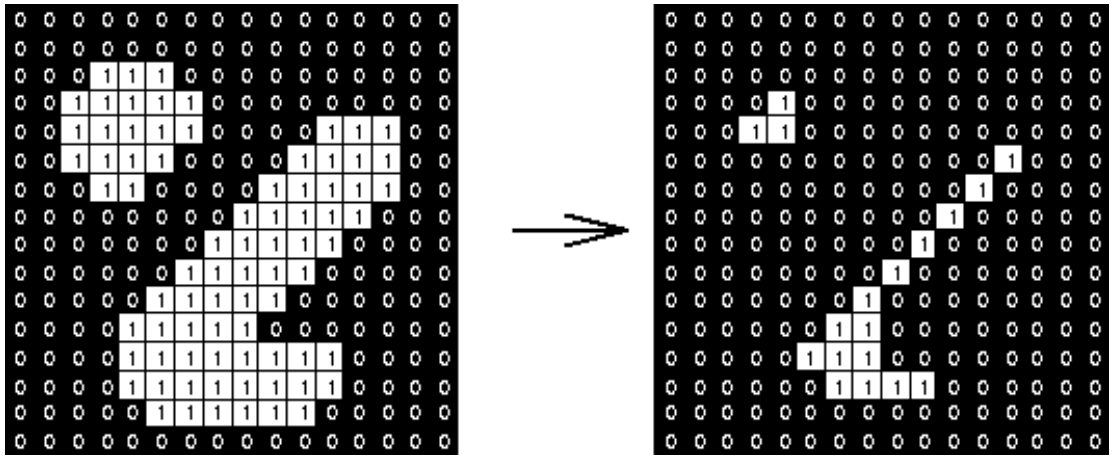
- Used generally on binary images, e.g., background subtraction results!
- Used on gray value images, if viewed as a stack to binary images.

Good for, e.g.,

- Noise removal in background
- Removal of holes in foreground / background

A first Example: Erosion

- **Erosion** is an important morphological operation



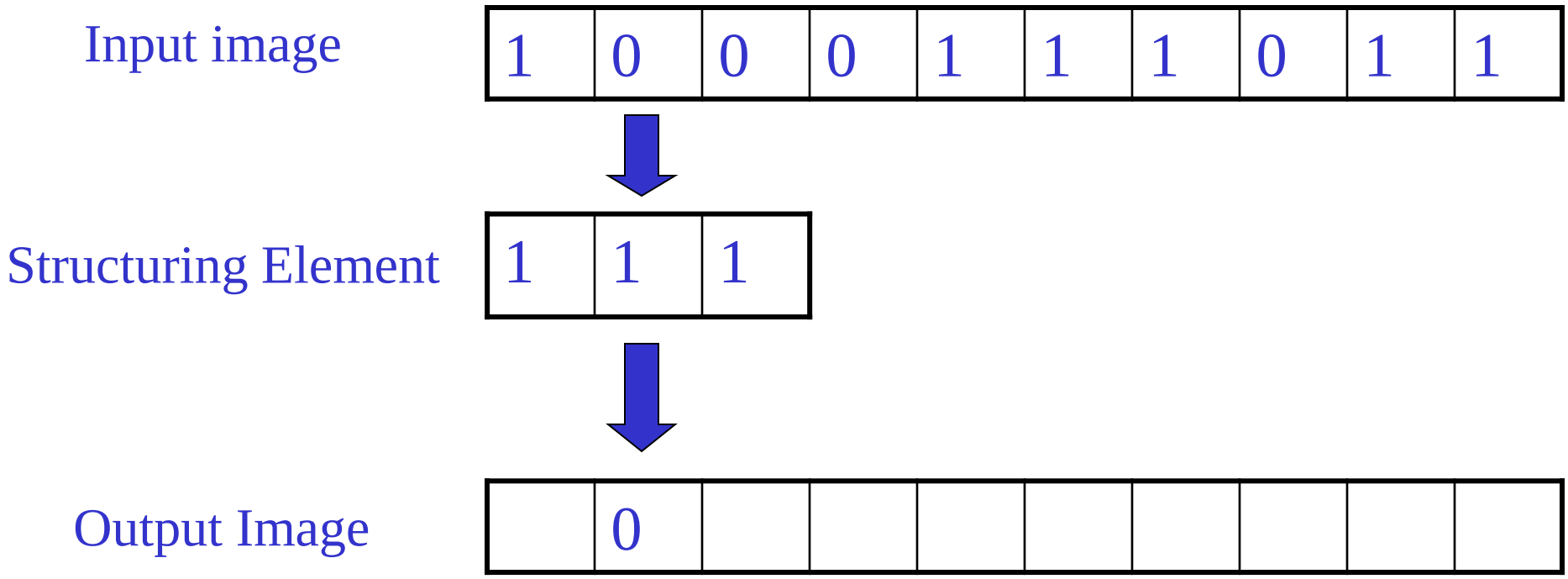
- Applied **Structuring Element**:

1	1	1
1	1	1
1	1	1

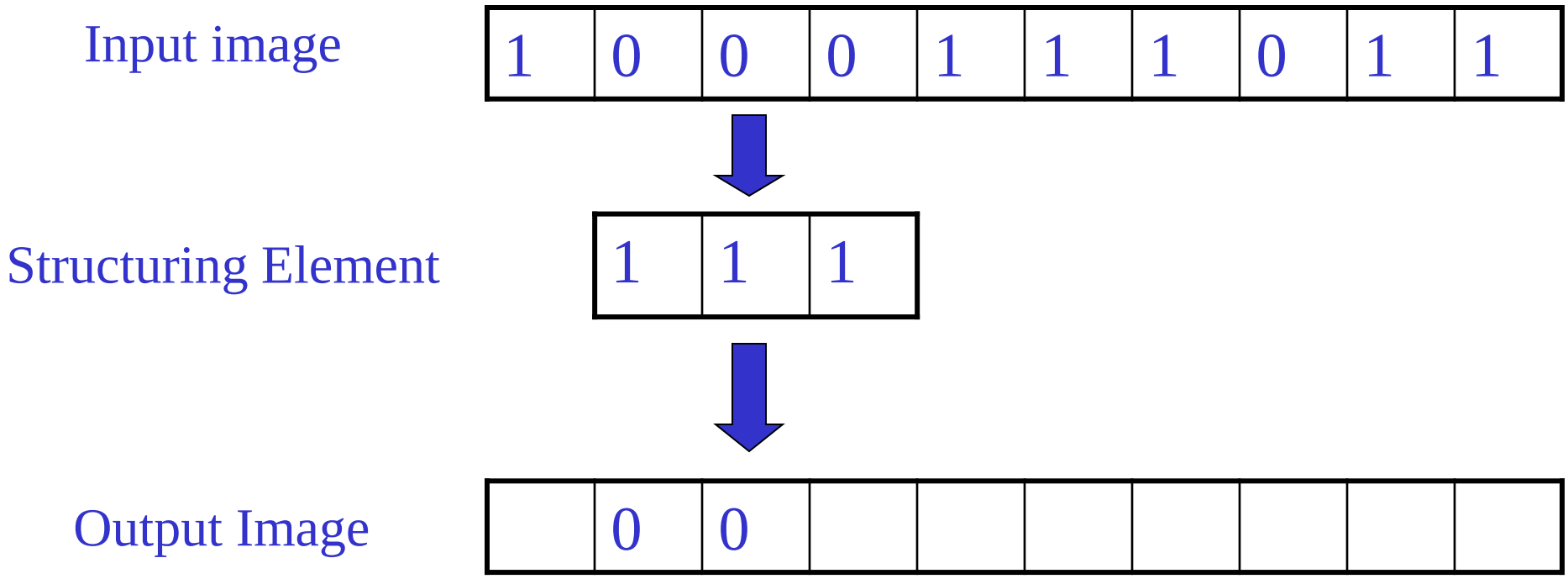
Set of coordinate points =

{ (-1, -1), (0, -1), (1, -1),
(-1, 0), (0, 0), (1, 0),
(-1, 1), (0, 1), (1, 1) }

Example for Erosion

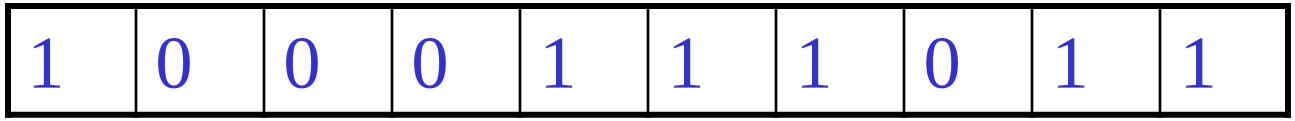


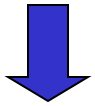
Example for Erosion



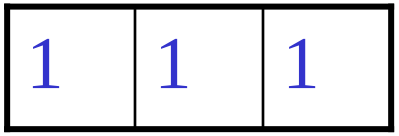
Example for Erosion

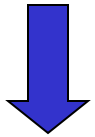
Input image



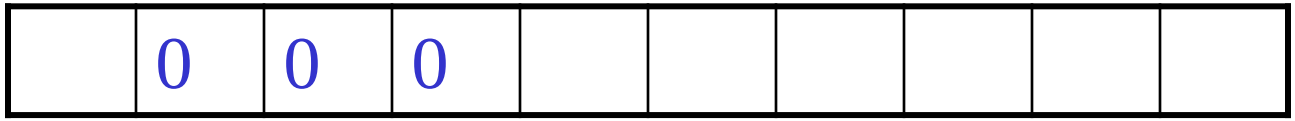


Structuring Element



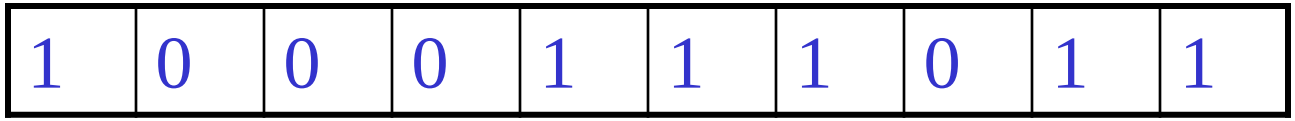


Output Image

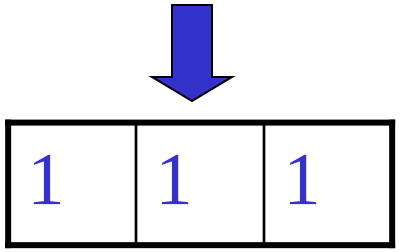


Example for Erosion

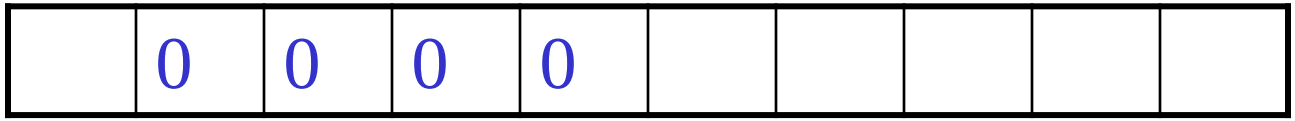
Input image



Structuring Element

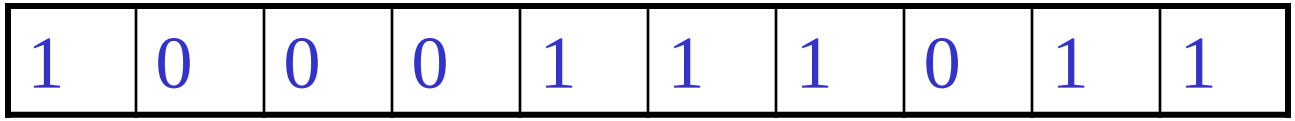


Output Image

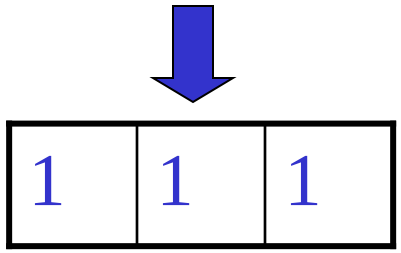


Example for Erosion

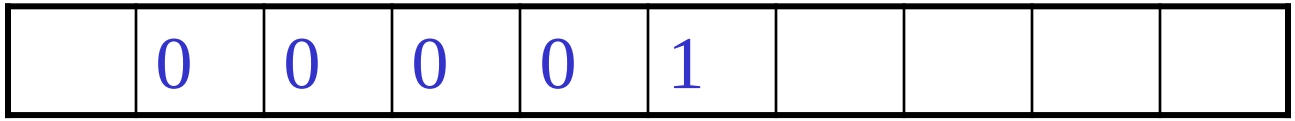
Input image



Structuring Element

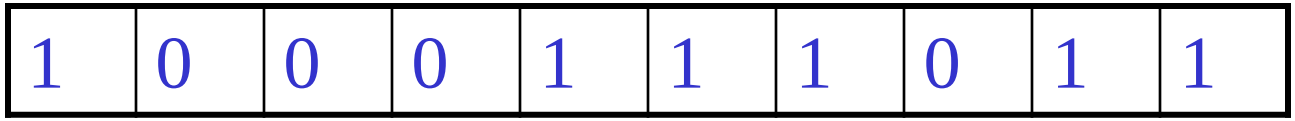


Output Image

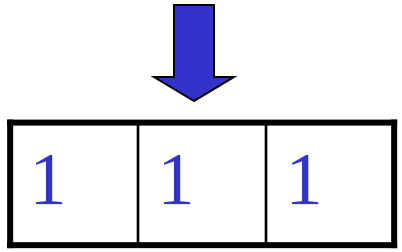


Example for Erosion

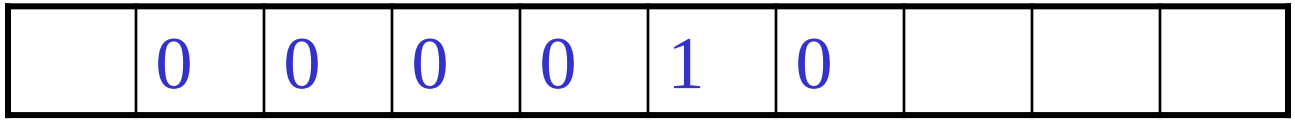
Input image



Structuring Element

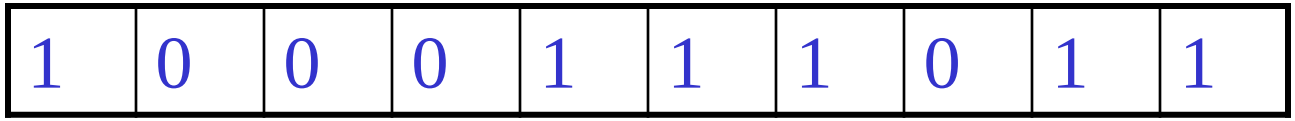


Output Image

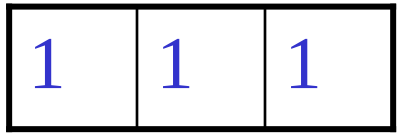


Example for Erosion

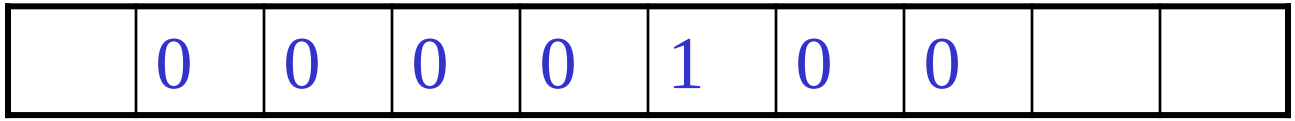
Input image



Structuring Element



Output Image



Example for Erosion

Input image

1	0	0	0	1	1	1	0	1	1

Structuring Element

1	1	1

Output Image

	0	0	0	0	1	0	0	0	

Introduction

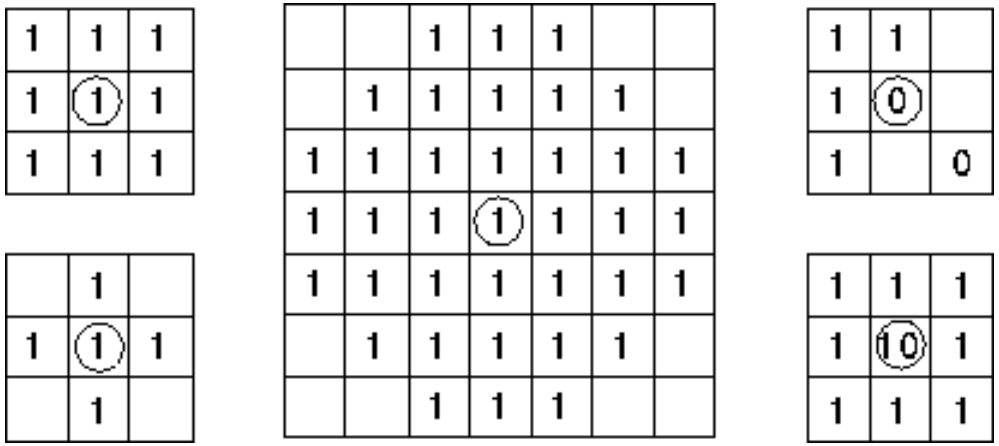
- Structuring Element
- Erosion
- Dilation
- Opening
- Closing
- Outlook: Hit-and-miss Operation, Thinning, Thickening

Structuring Element (Kernel)

- Structuring Elements can have varying sizes
- Usually, element values are 0,1 and none(!)
- Structural Elements have an origin
- For thinning, other values are possible
- Empty spots in the Structuring Elements are *don't care*'s!

Box →

Disc →



Examples of stucturing elements

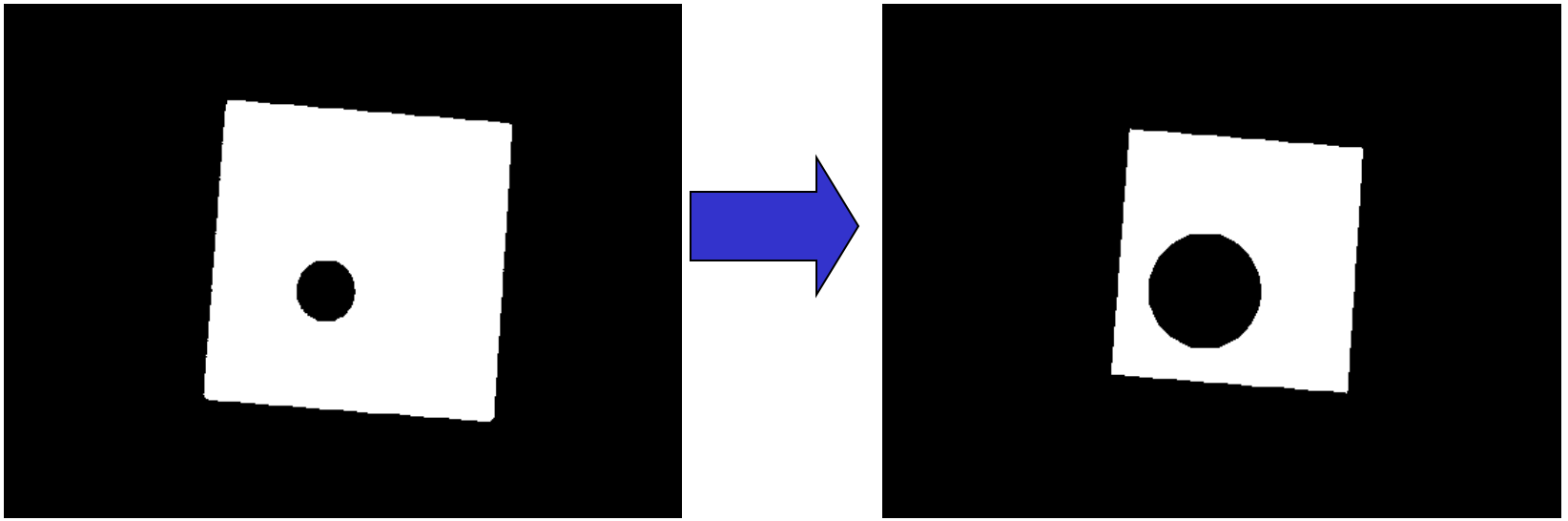
Dilation & Erosion

- Basic operations
- Are dual to each other:
  - Erosion shrinks foreground, enlarges Background
  - Dilation enlarges foreground, shrinks background

Erosion

- **Erosion** is the set of all points in the image, where the structuring element “fits into”.
- Consider each foreground pixel in the input image
  - If the structuring element fits in, write a “1” at the origin of the structuring element!
- Simple application of **pattern matching**
- **Input:**
  - **Binary Image (Gray value)**
  - **Structuring Element, containing only 1s!**

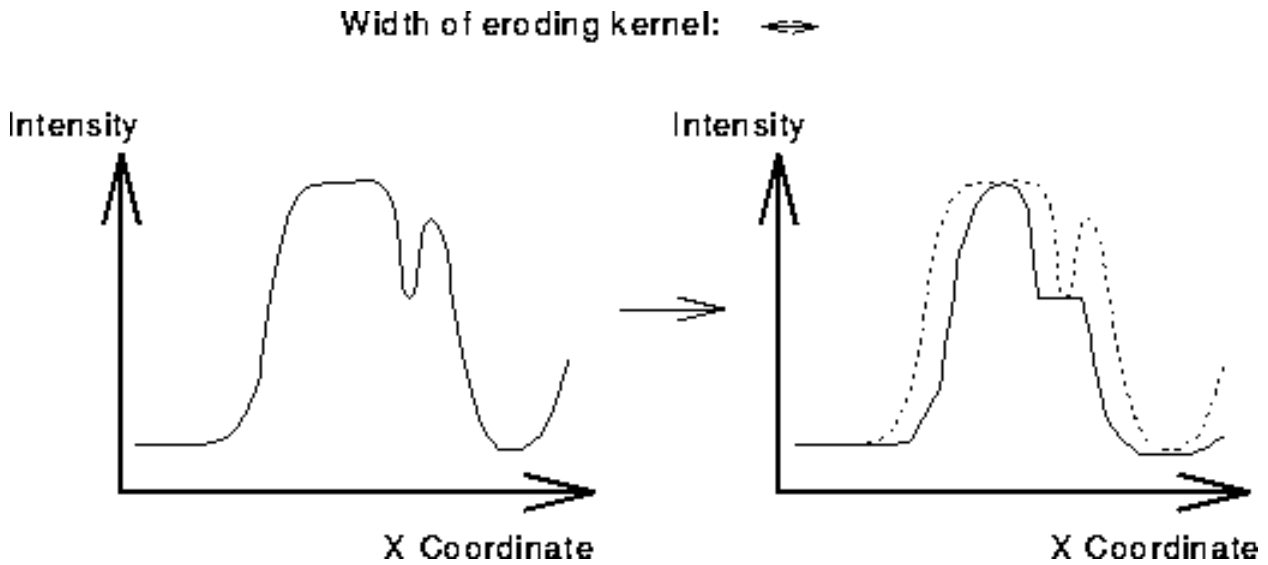
Another example of erosion



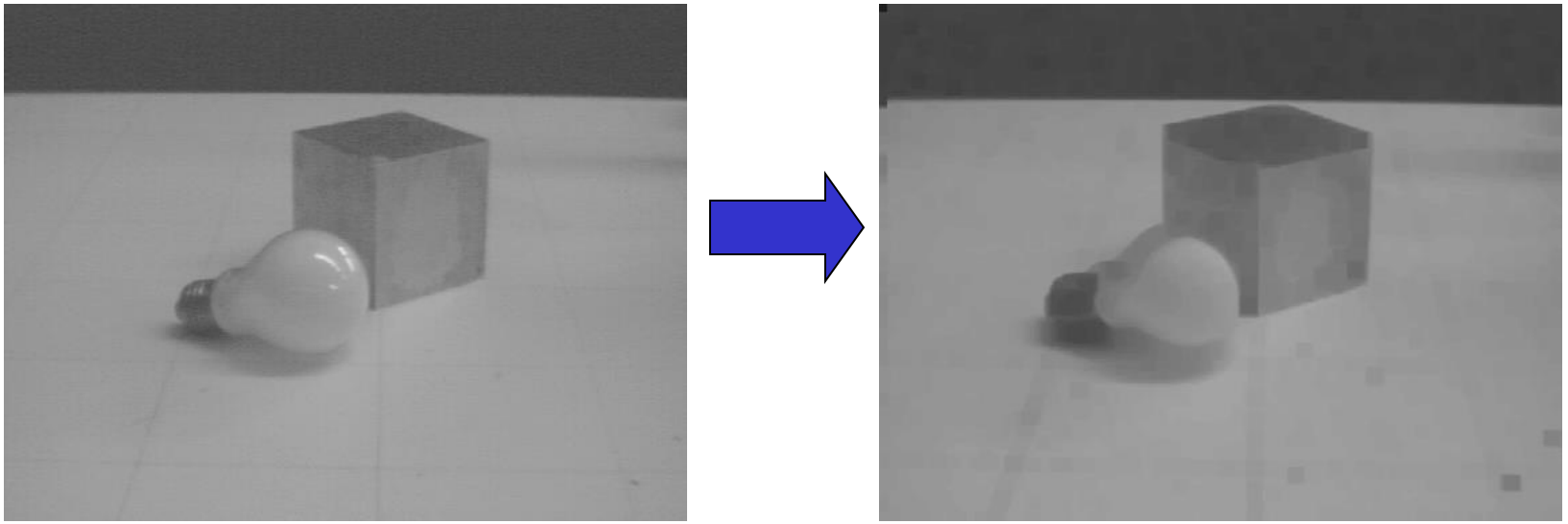
- White = 0, black = 1, dual property, image as a result of erosion gets darker

Erosion on Gray Value Images

- View gray value images as a stack of binary images!



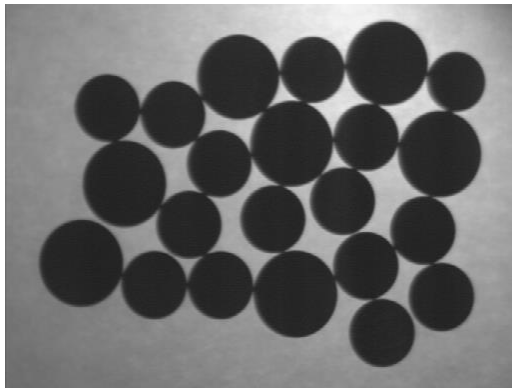
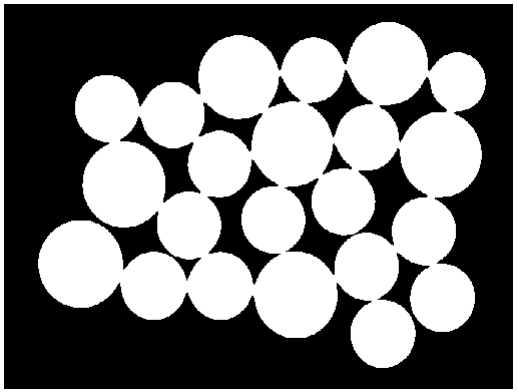
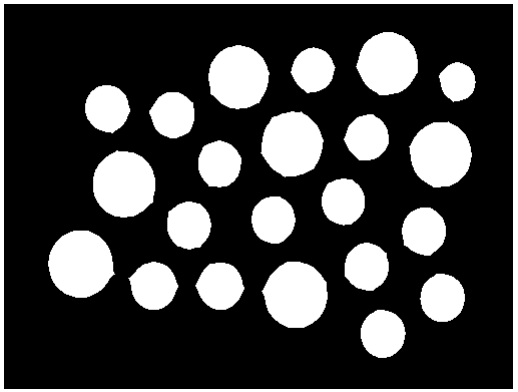
Erosion on Gray Value Images



- Images get darker!

Counting Coins

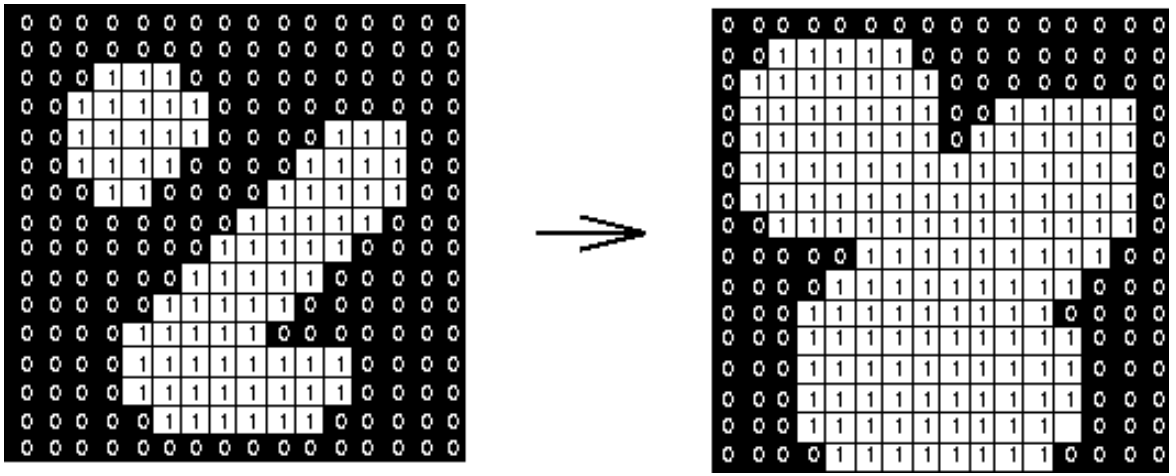
- Counting coins is difficult because they touch each other!
- Solution: Binarization and Erosion separates them!

DILATION

Example: Dilation

- **Dilation** is an important morphological operation



- Applied **Structuring Element**:

1	1	1
1	1	1
1	1	1

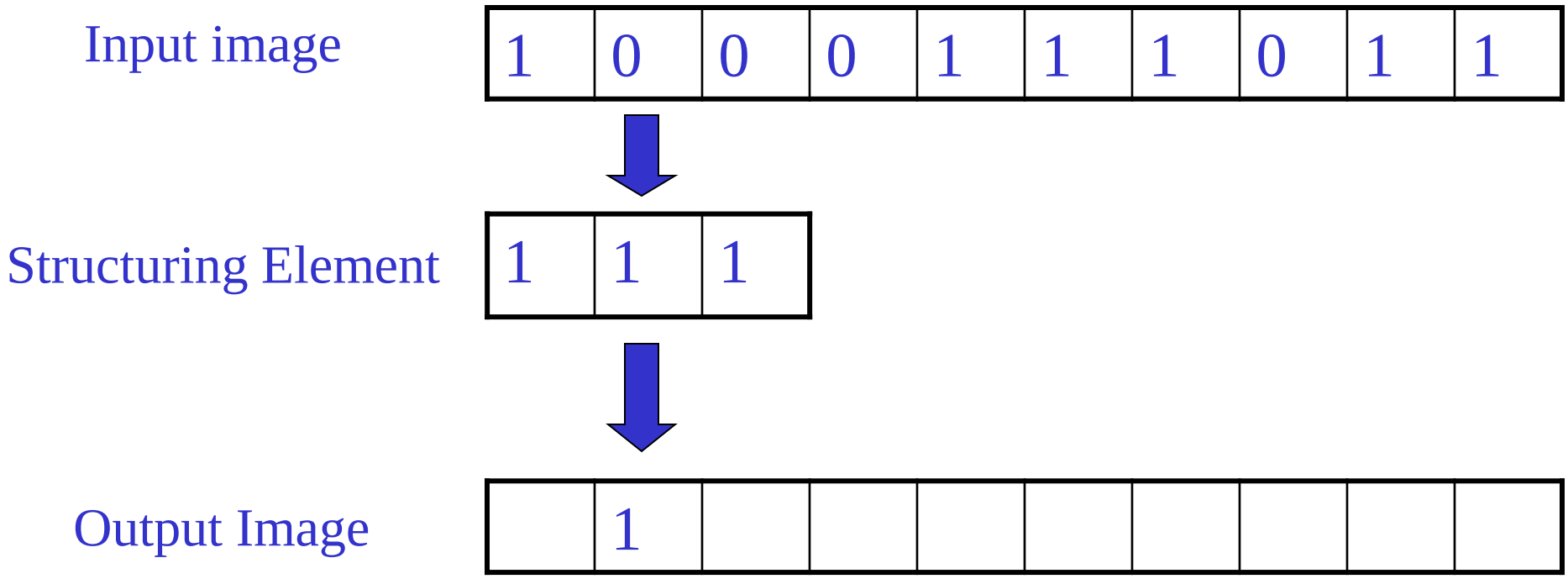
Set of coordinate points =

{ (-1, -1), (0, -1), (1, -1),
(-1, 0), (0, 0), (1, 0),
(-1, 1), (0, 1), (1, 1) }

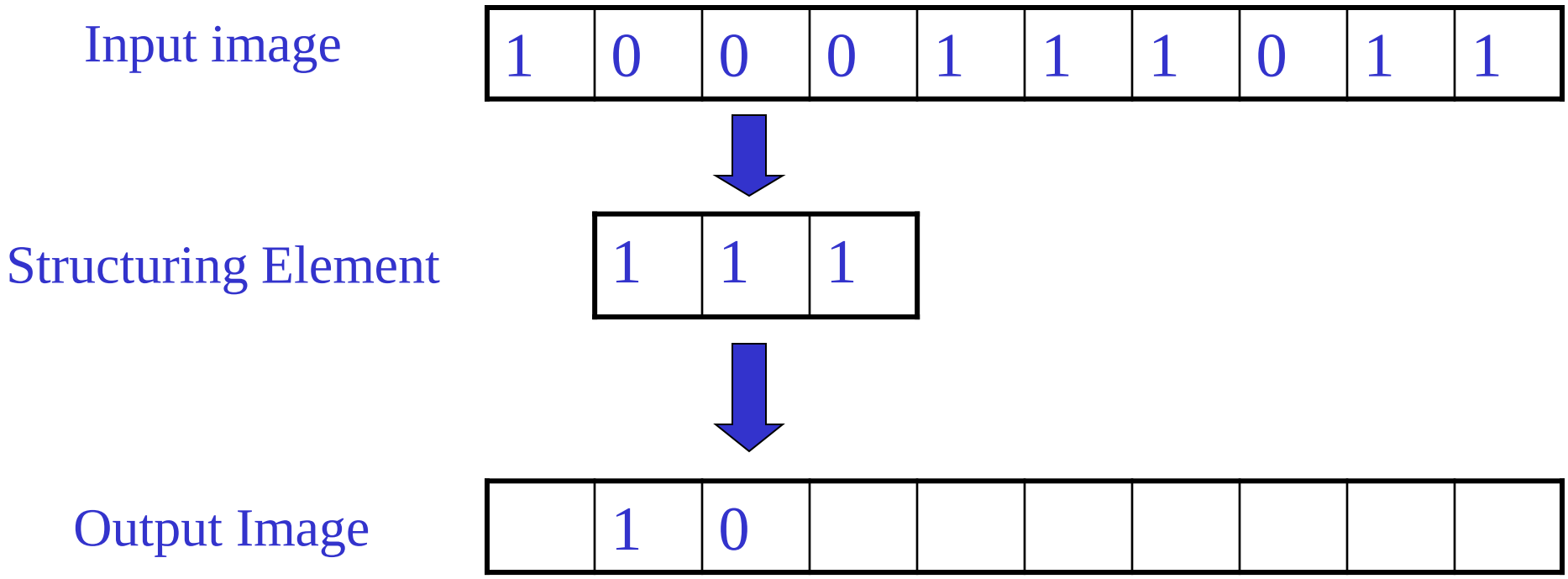
Dilation

- **Dilation** is the set of all points in the image, where the structuring element “touches” the foreground.
- Consider each pixel in the input image
  - If the structuring element touches the foreground image, write a “1” at the origin of the structuring element!
- **Input:**
  - **Binary Image**
  - **Structuring Element, containing only 1s!!**

Example for Dilation

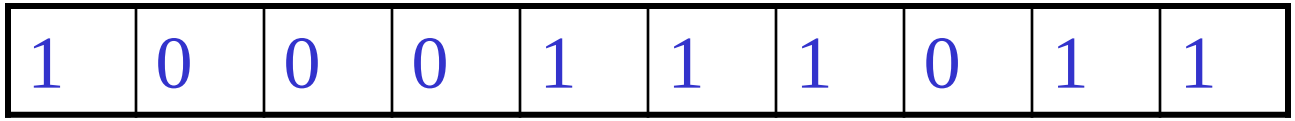


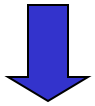
Example for Dilation



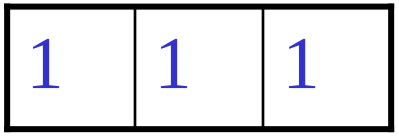
Example for Dilation

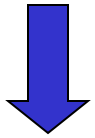
Input image



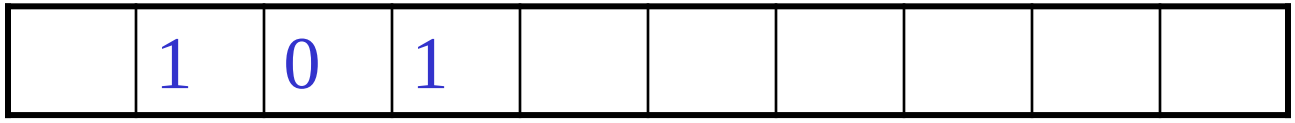


Structuring Element



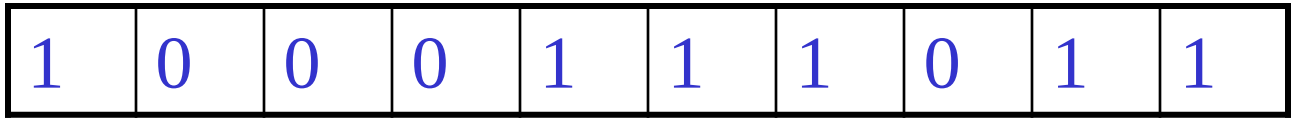


Output Image

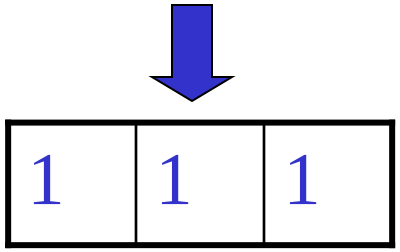


Example for Dilation

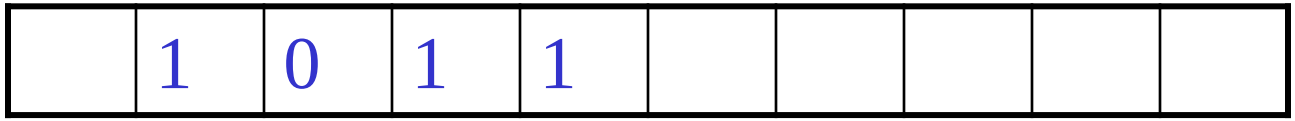
Input image



Structuring Element

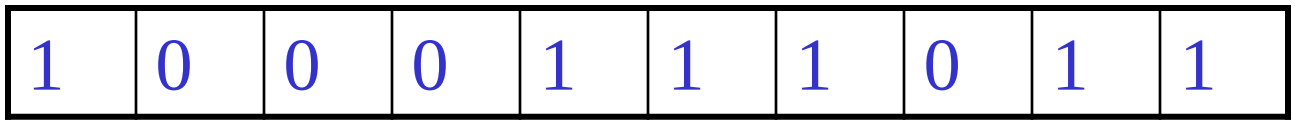


Output Image

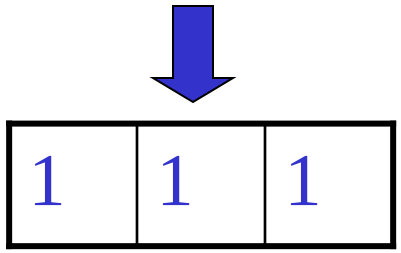


Example for Dilation

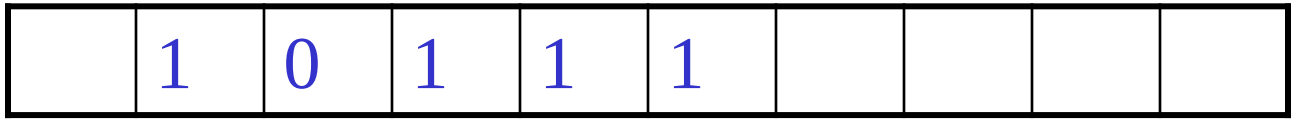
Input image



Structuring Element

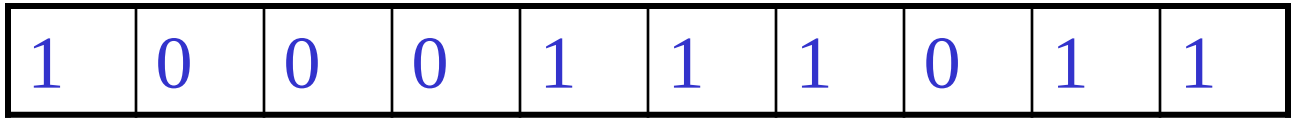


Output Image

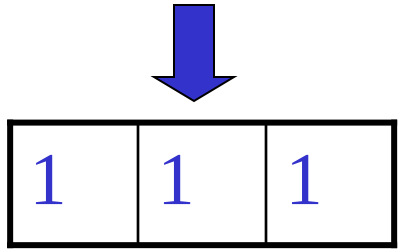


Example for Dilation

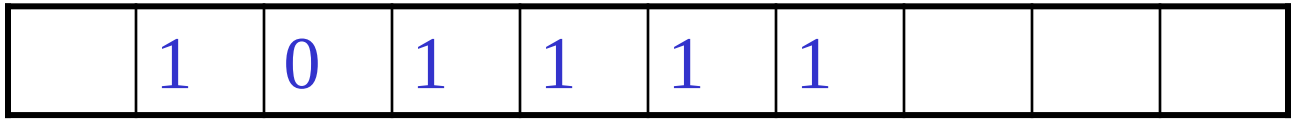
Input image



Structuring Element

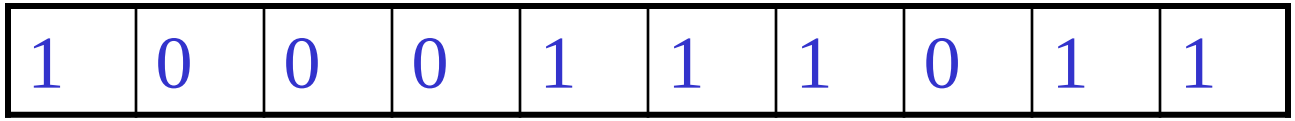


Output Image

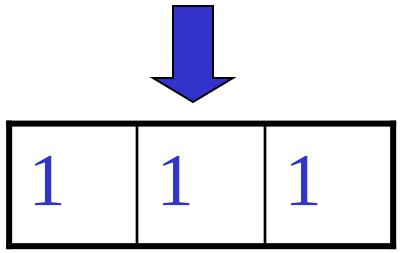


Example for Dilation

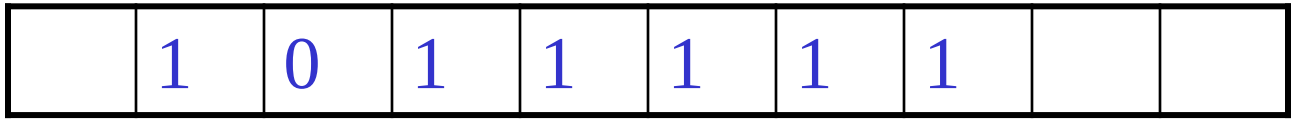
Input image



Structuring Element



Output Image



Example for Dilation

Input image

1	0	0	0	1	1	1	0	1	1

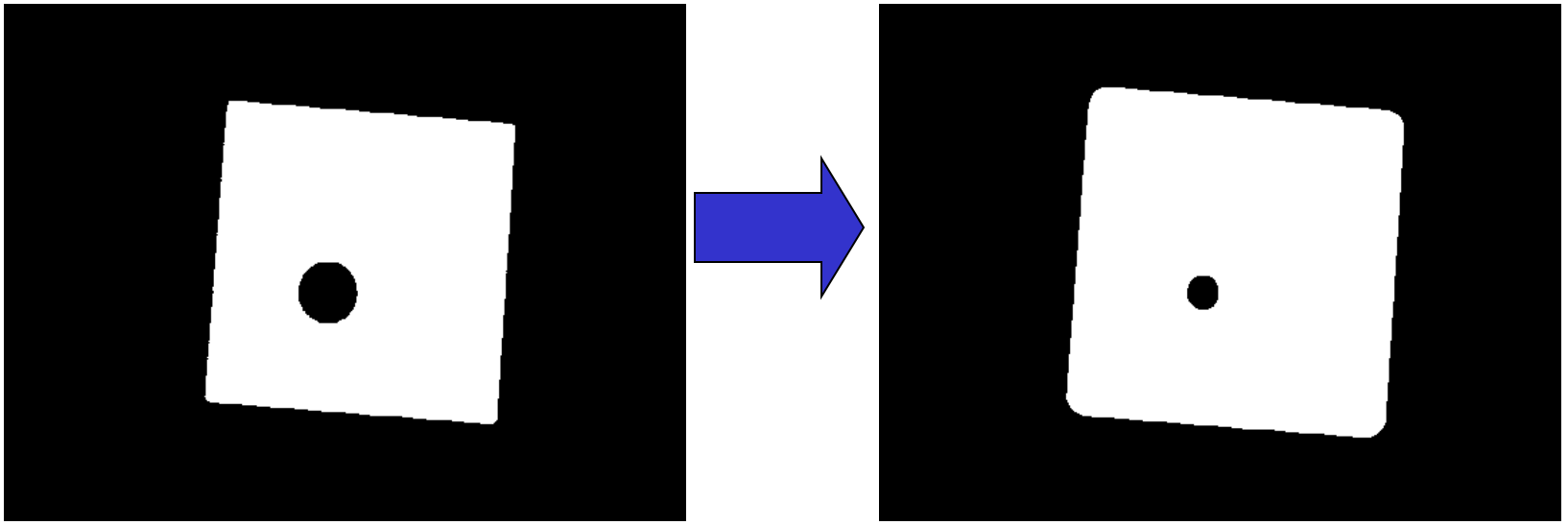
Structuring Element

1	1	1

Output Image

	1	0	1	1	1	1	1	1	

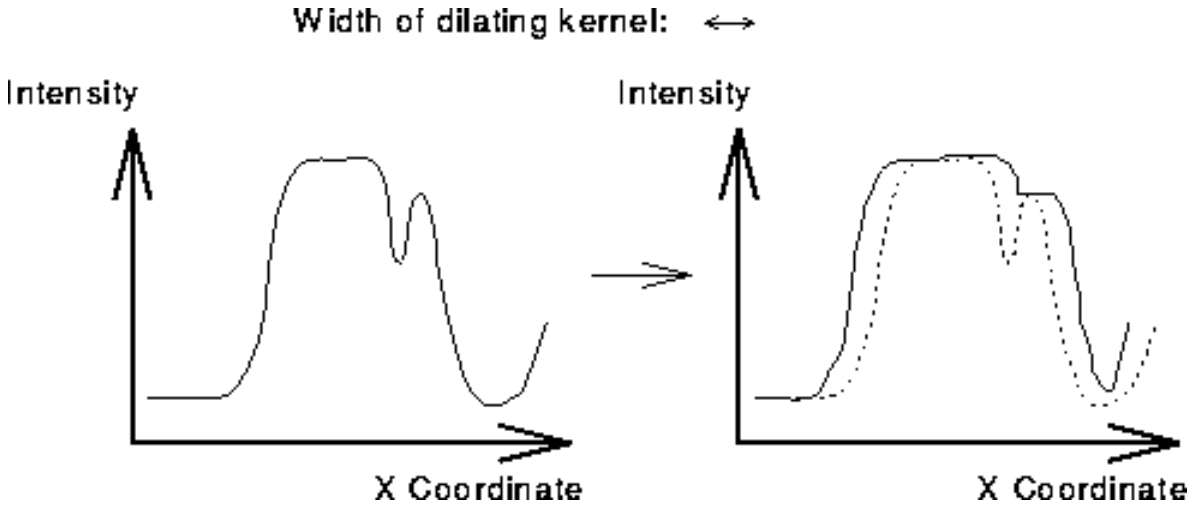
Another Dilation Example



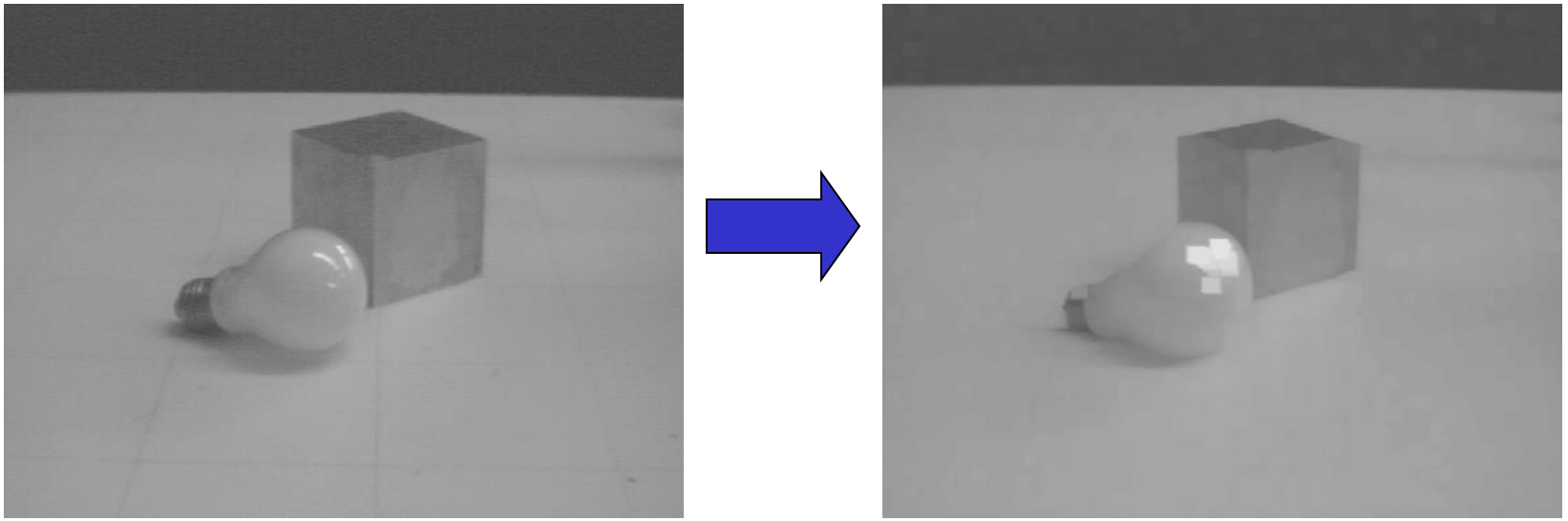
- Image get lighter, more uniform intensity

Dilation on Gray Value Images

- View gray value images as a stack of binary images!



Dilation on Gray Value Images

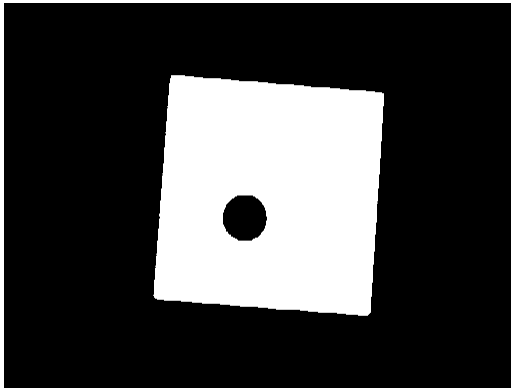
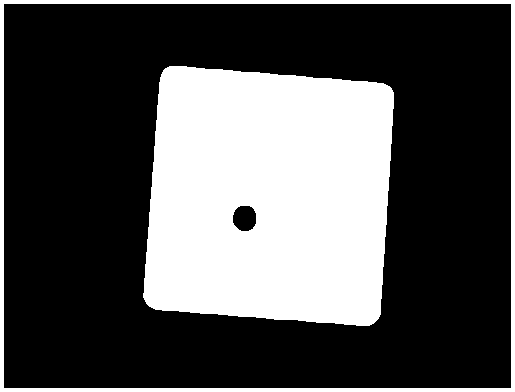
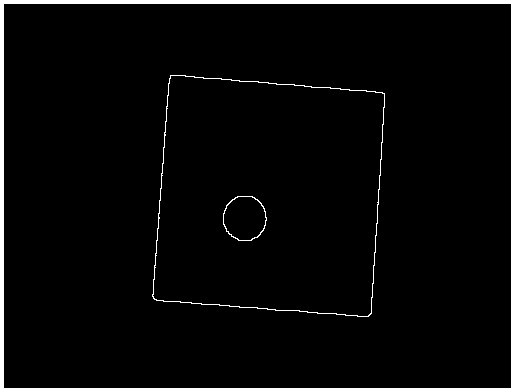


- More uniform intensity

Edge Detection

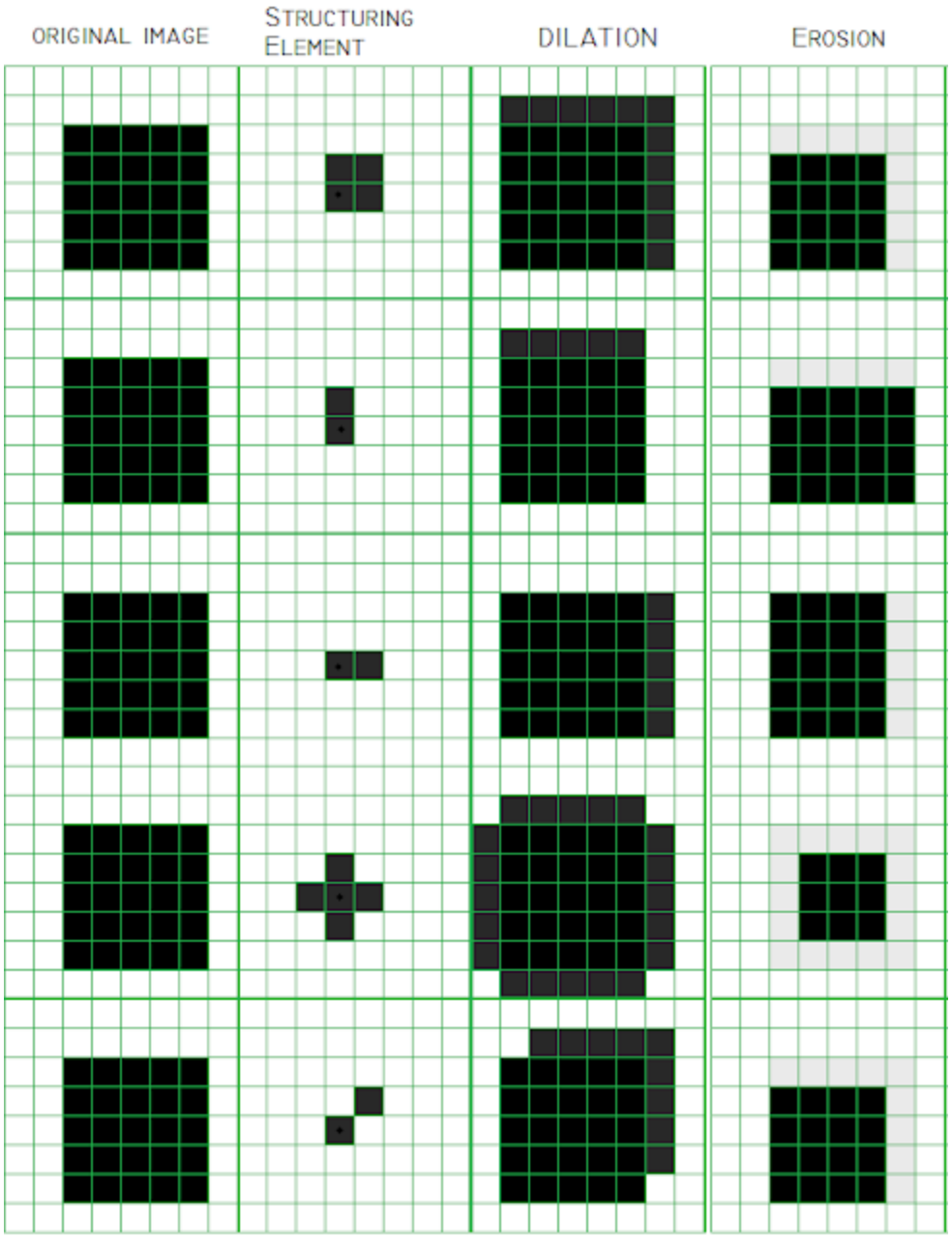
- Edge Detection

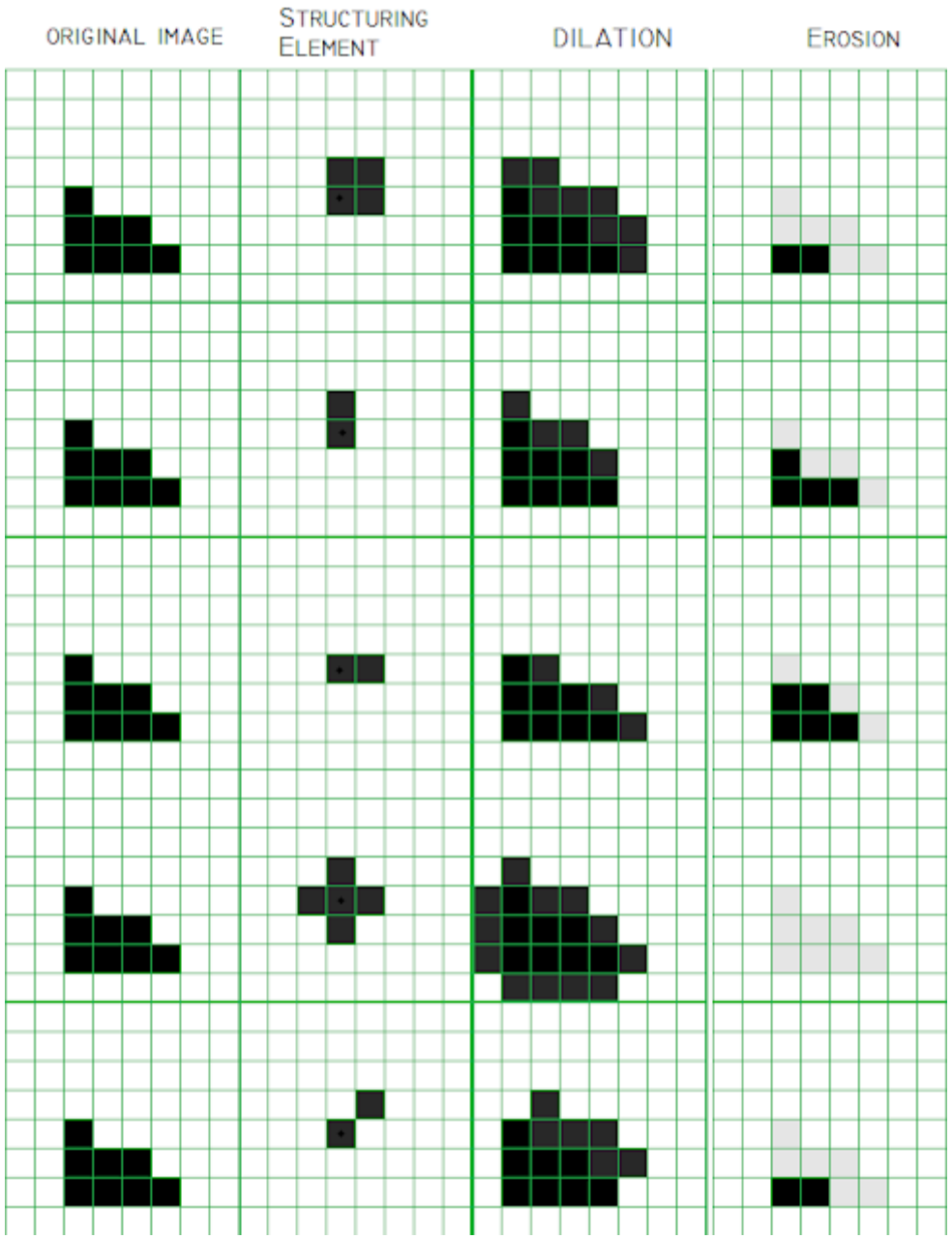
1. Dilate input image
2. Subtract input image from dilated image
3. Edges remain!

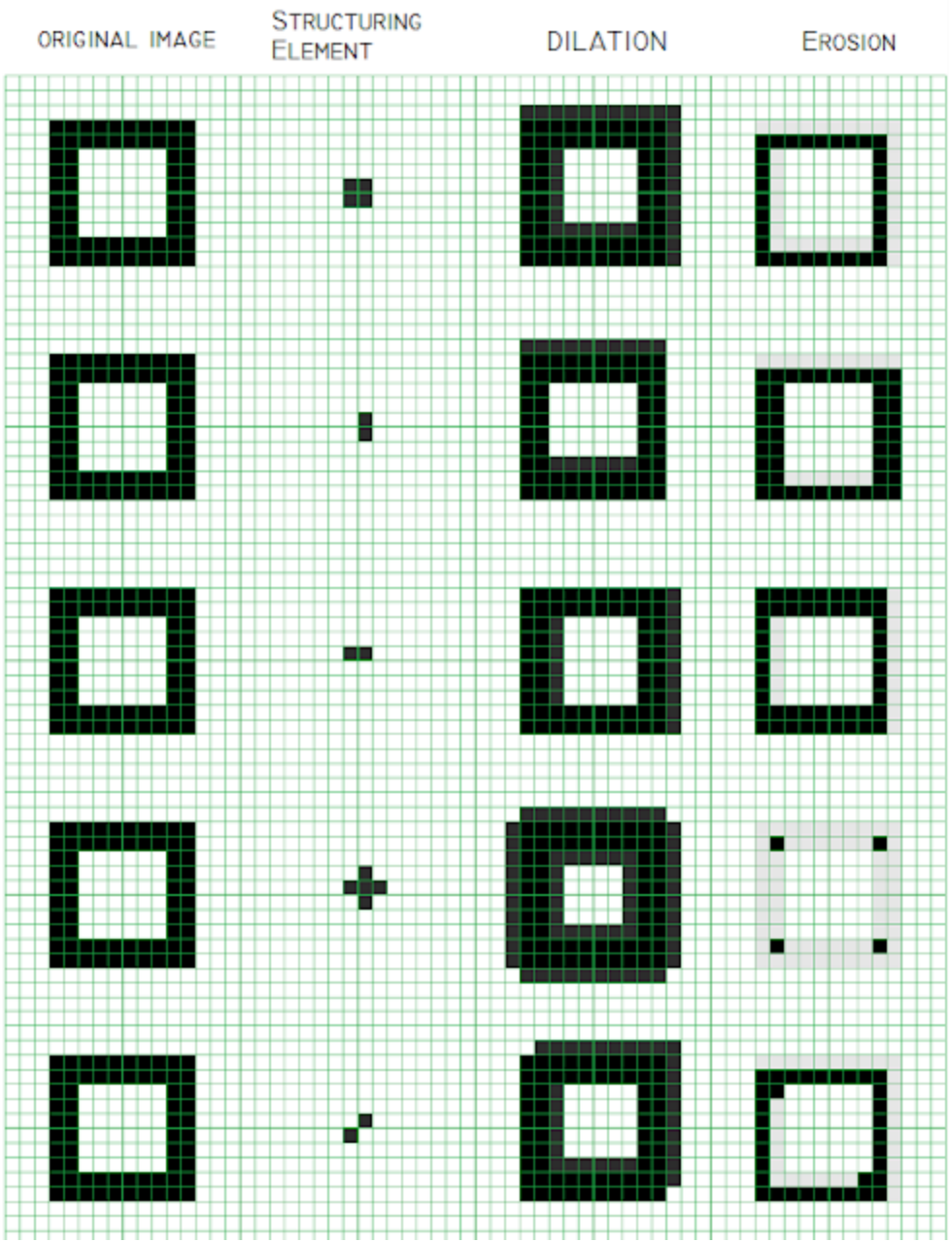




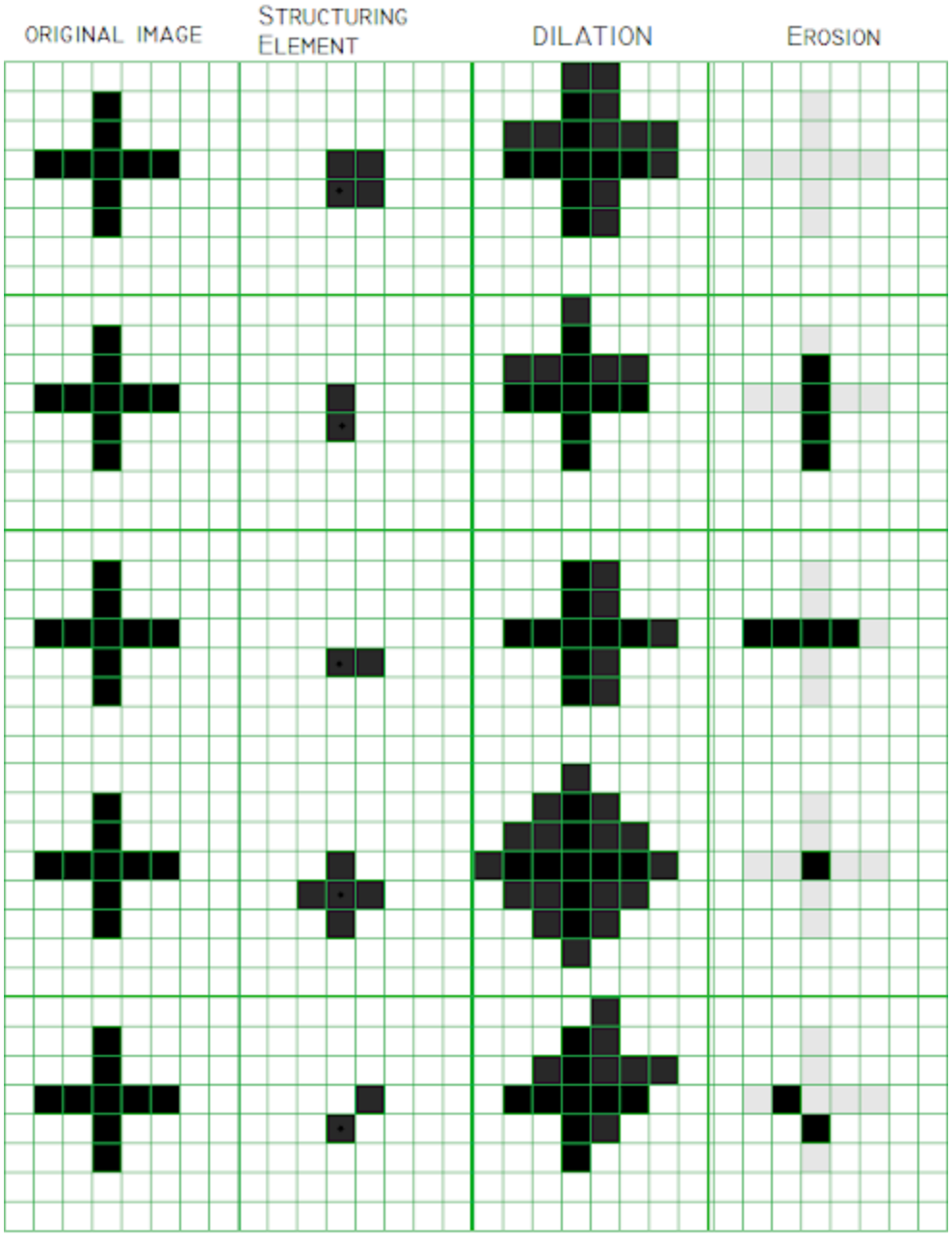
Opening & Closing

- Important operations
- Derived from the fundamental operations
  - Dilatation
  - Erosion
- Usually applied to binary images, but gray value images are also possible
- Opening and closing are dual operations


ORIGINAL IMAGE
STRUCTURING ELEMENT
DILATION
EROSION


ORIGINAL IMAGE
STRUCTURING ELEMENT
DILATION
EROSION


ORIGINAL IMAGE
STRUCTURING ELEMENT
DILATION
EROSION


ORIGINAL IMAGE
STRUCTURING ELEMENT
DILATION
EROSION

OPENING

Opening

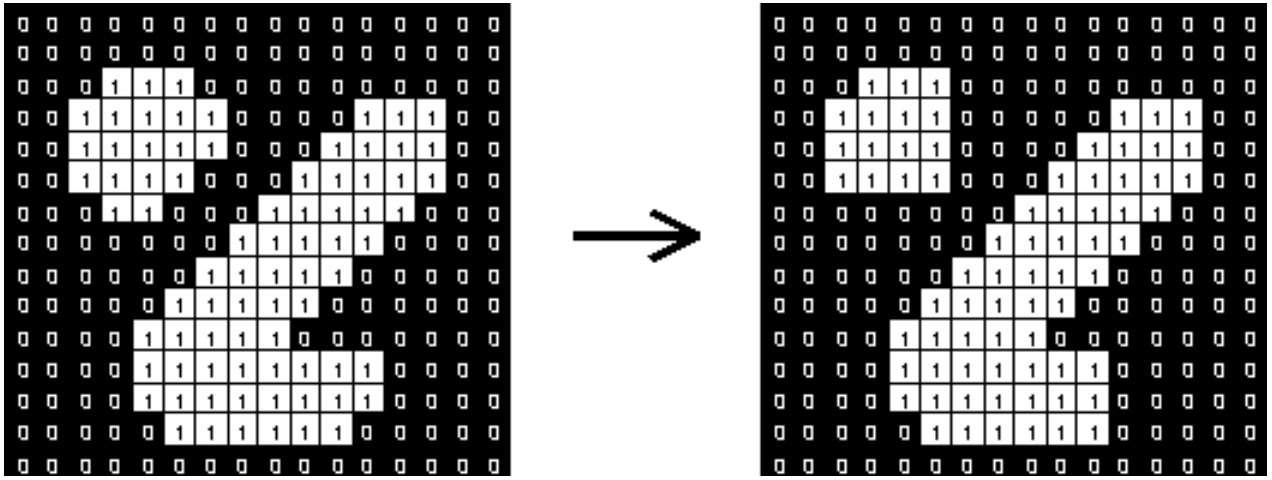
- Similar to Erosion
  - Spot and noise removal
  - Less destructive
- Erosion next dilation
- the s*ame structuring element for both operations*.
- Input:
  - Binary Image
  - Structuring Element, containing only 1s!

Opening

- Take the structuring element (SE) and slide it around *inside* each foreground region.
  - All pixels which can be covered by the SE with the SE being entirely within the foreground region will be preserved.
  - All foreground pixels which can *not* be reached by the structuring element without lapping over the edge of the foreground object will be eroded away!
- Opening is **idempotent:** Repeated application has no further effects!

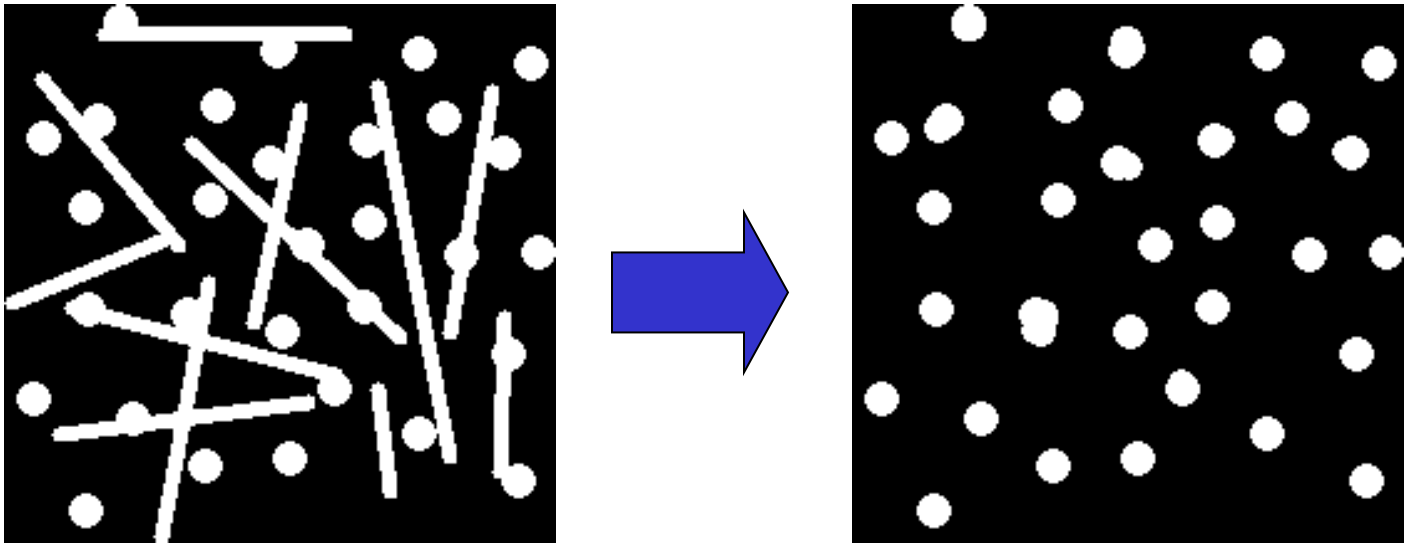
Opening

- Structuring element: 3x3 square



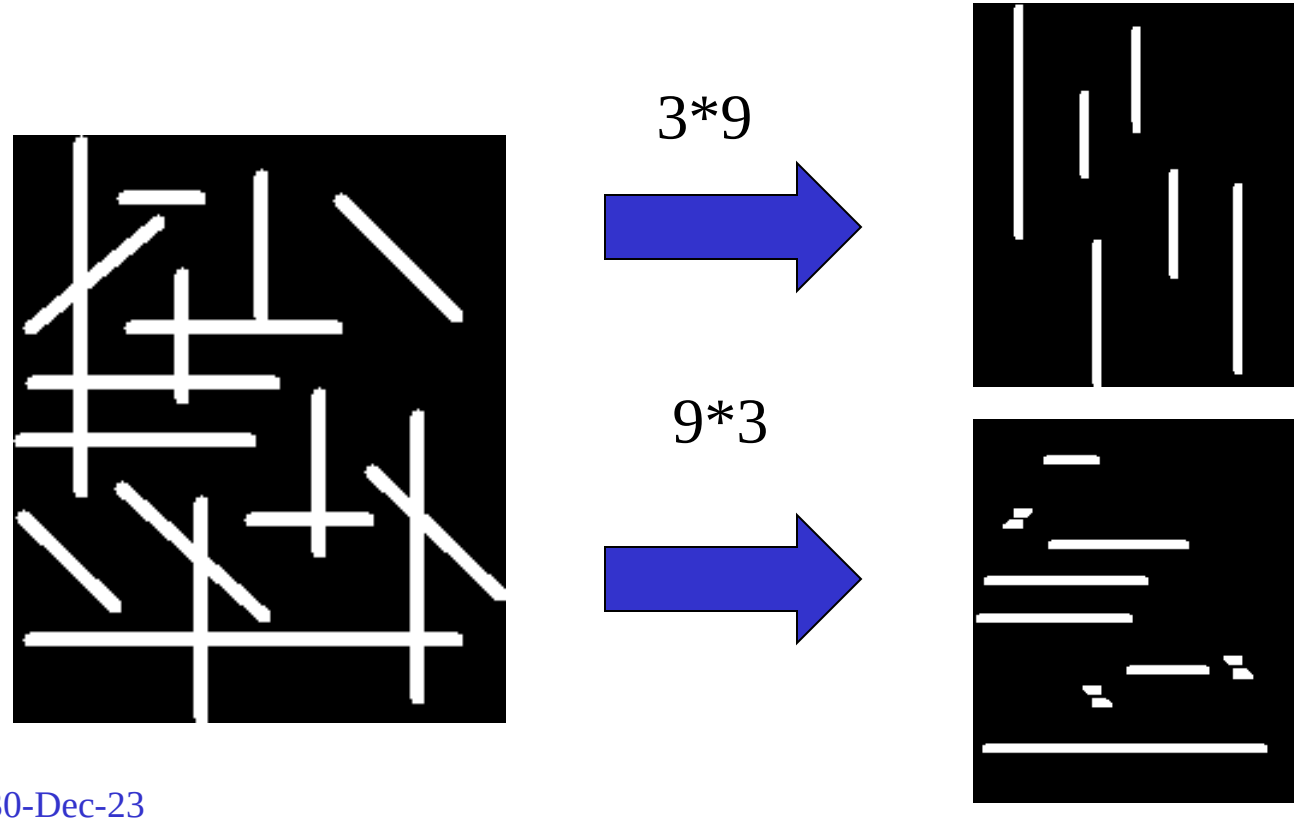
Opening Example

- Opening with a 11 pixel diameter disc



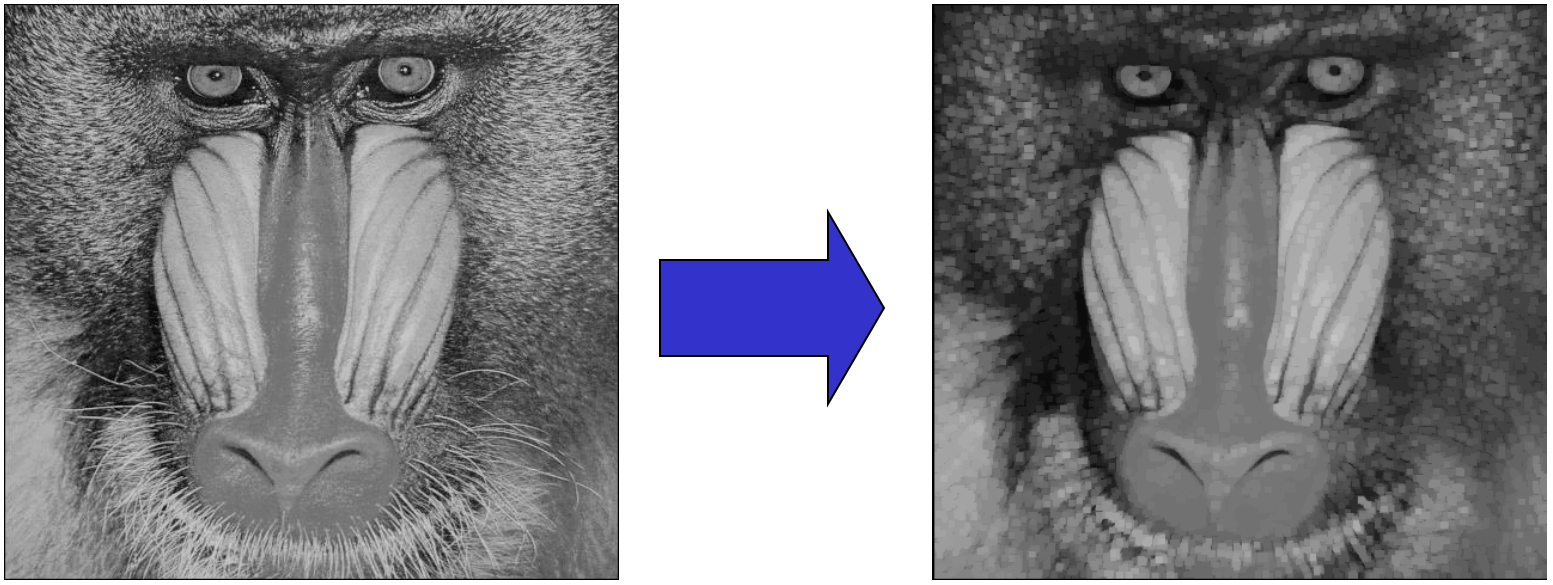
Opening Example

- 3x9 and 9x3 Structuring Element



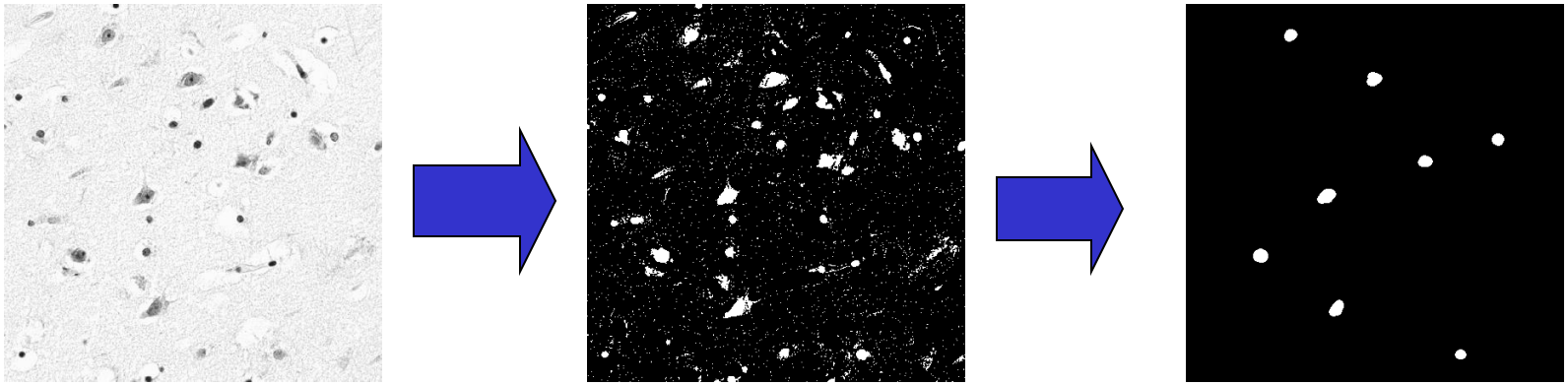
Opening on Gray Value Images

- 5x5 square structuring element



Use Opening for Separating Blobs

- Use large structuring element that fits into the big blobs
- Structuring Element: 11 pixel disc



CLOSING

Closing

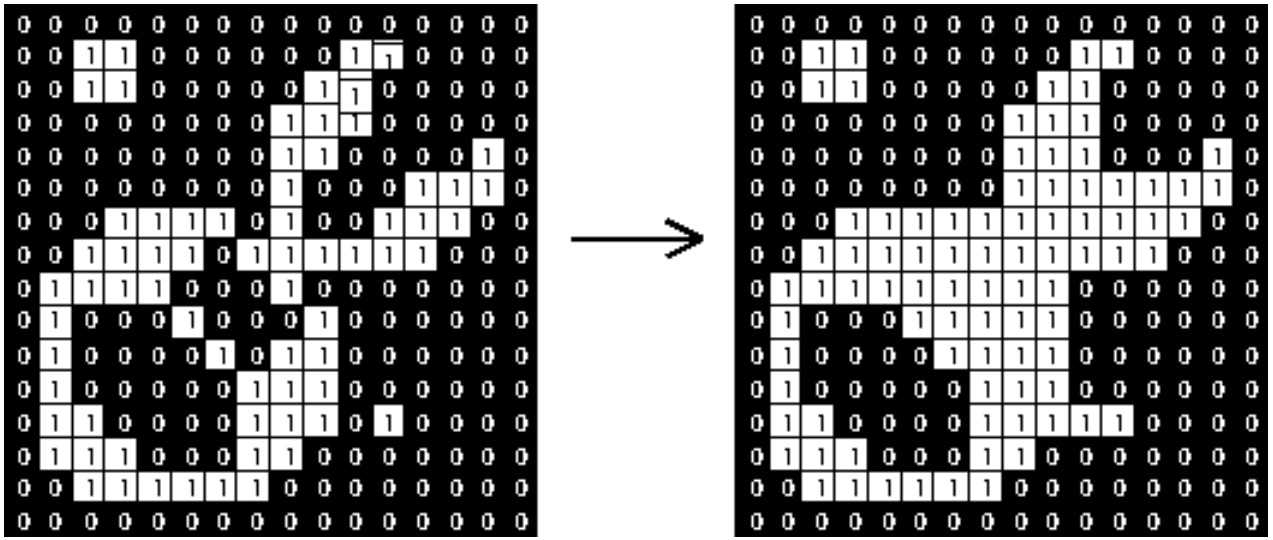
- Similar to Dilation
  - Removal of holes
  - Tends to enlarge regions, shrink background
- Closing is defined as a Dilatation, followed by an Erosion *using the same structuring element for both operations.*
- Dilation next erosion!
- Input:
  - Binary Image
  - Structuring Element, containing only 1s!

Closing

- Take the structuring element (SE) and slide it around *outside* each foreground region.
  - All background pixels which can be covered by the SE with the SE being entirely within the background region will be preserved.
  - All background pixels which can *not* be reached by the structuring element without lapping over the edge of the foreground object will be turned into a foreground.
- Opening is **idempotent:** Repeated application has no further effects!

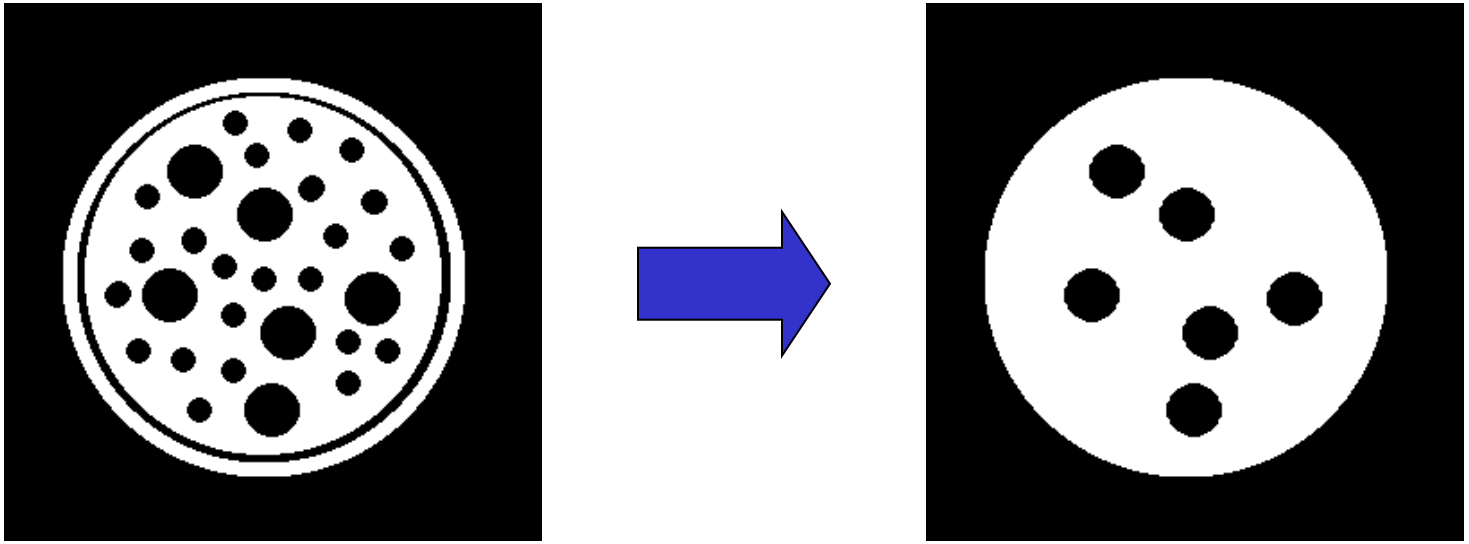
Closing

- Structuring element: 3x3 square



Closing Example

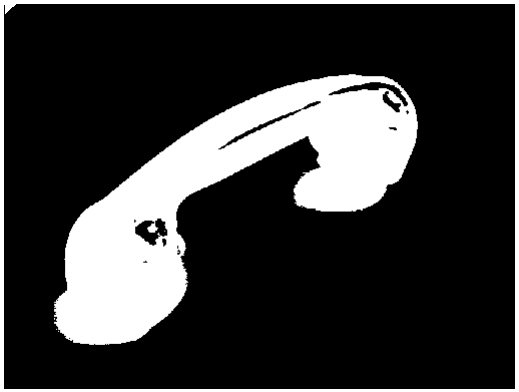
- Closing operation with a 22 pixel disc
- Closes small holes in the foreground

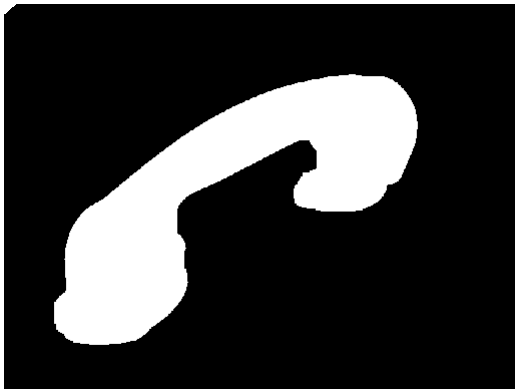


Closing Example 1

1. Threshold
2. Closing with disc of size 20



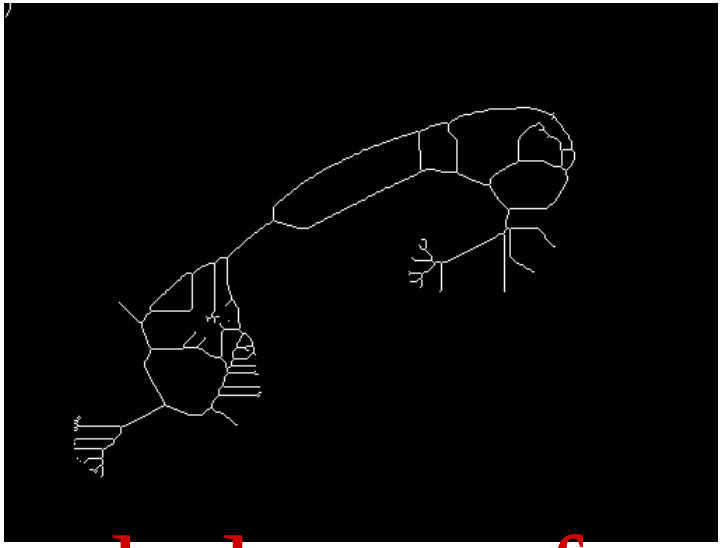




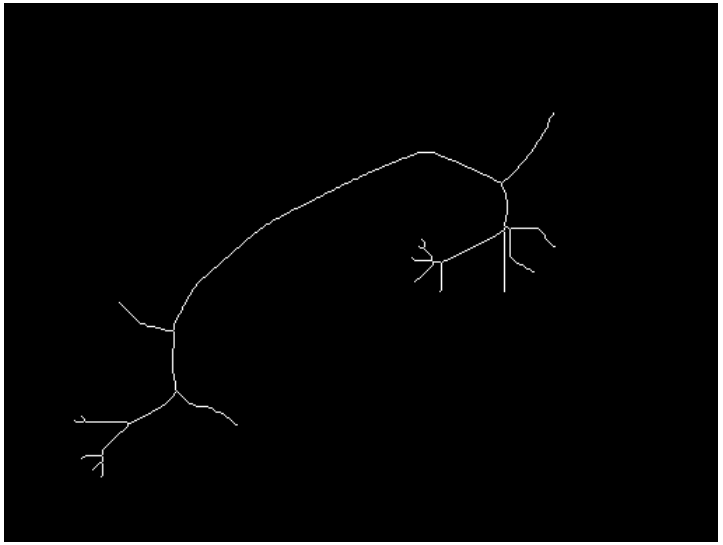
Thresholded closed

Closing Example 2

- Good for further processing: E.g. Skeleton operation looks better for closed image!



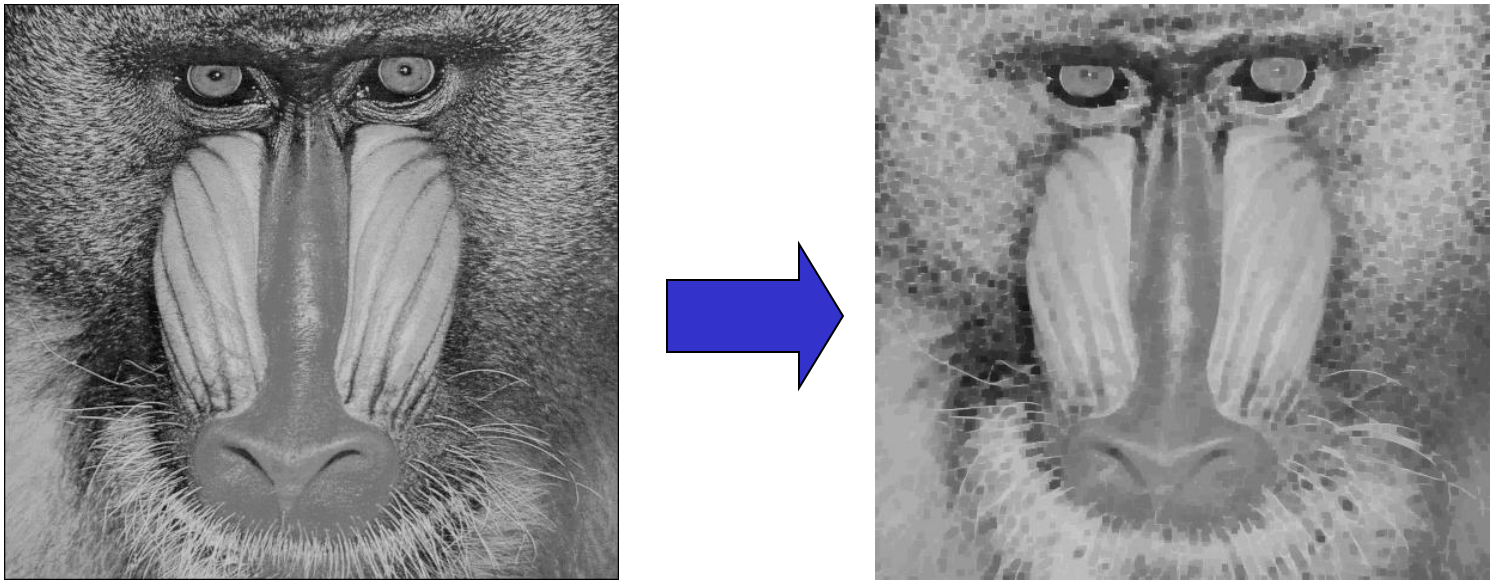
skeleton of Thresholded



skeleton of Thresholded and next closed

Closing Gray Value Images

- 5x5 square structuring element



Opening & Closing

- Opening is the *dual* of closing
- *i.e.* opening the foreground pixels with a particular structuring element
- is equivalent to closing the background pixels with the same element.

Application



HIT and MISS

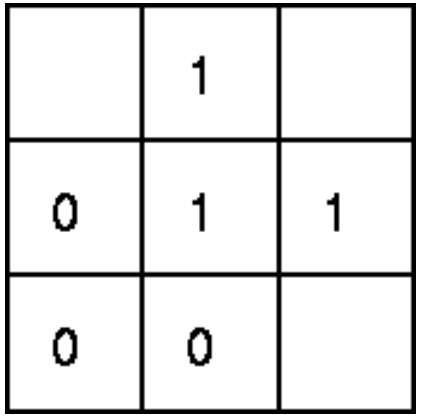
Hit-and-miss Transform

- Used to look for particular patterns of foreground and background pixels

- Very simple object recognition
- All other morphological operations can be derived from it!!
- Input:
  - Binary Image
  - Structuring Element, containing 0s and 1s!!

Hit-and-miss Transform

- Example for a Hit-and-miss Structuring Element
- Contains 0s, 1s and *don't care*'s.
- Usually a "1" at the origin!

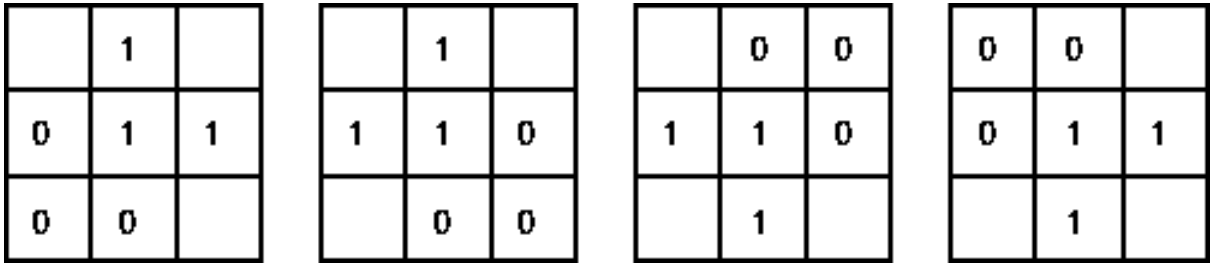


Hit-and-miss Transform

- Similar to Pattern Matching:
- If foreground and background pixels in the structuring element *exactly match* foreground and background pixels in the image, then the pixel underneath the origin of the structuring element is set to the foreground color.

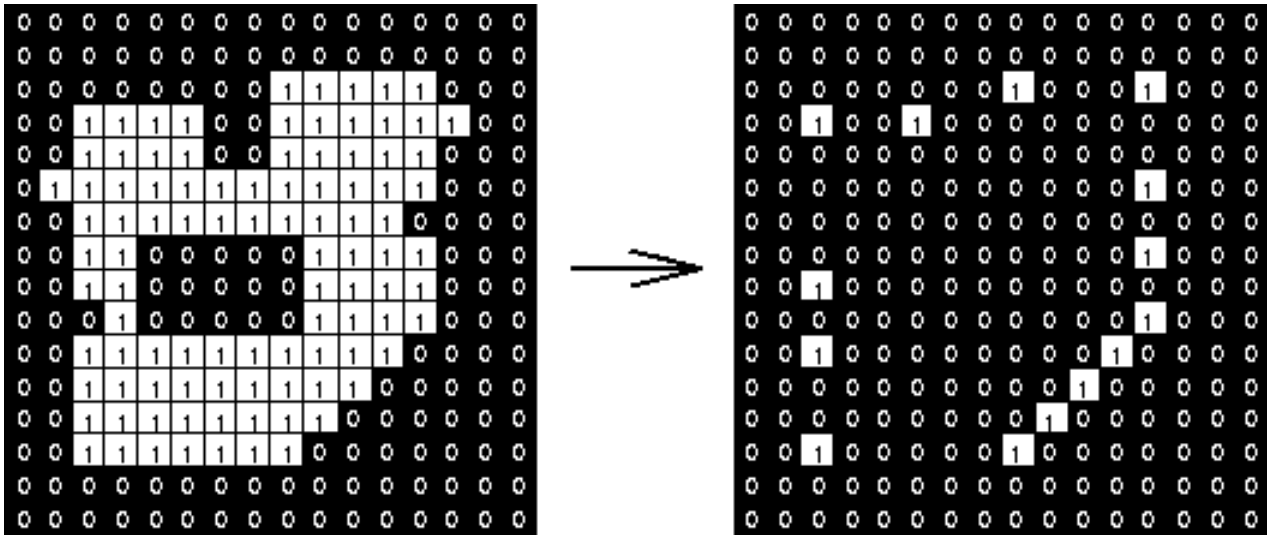
Corner Detection with Hit-and-miss Transform

- Structuring Elements representing four corners



Corner Detection with Hit-and-miss Transform

- Apply each Structuring Element
- Use OR operation to combine the four results



Basic THINNING

Thinning

1. Used to remove selected foreground pixels from binary images

2. After edge detection, lines are often thicker than one pixel.

3. Thinning can be used to thin those line to one pixel width.

Definition of Thinning

- Let ***K*** be a kernel and ***I*** be an image

$$\text{thin}\left(I,K\right)=I-\text{HitAndMiss}\ \left(I,K\right)$$

with 0-1=0!!

- If foreground and background fit the structuring element exactly, then the pixel at the origin of the SE is set to 0

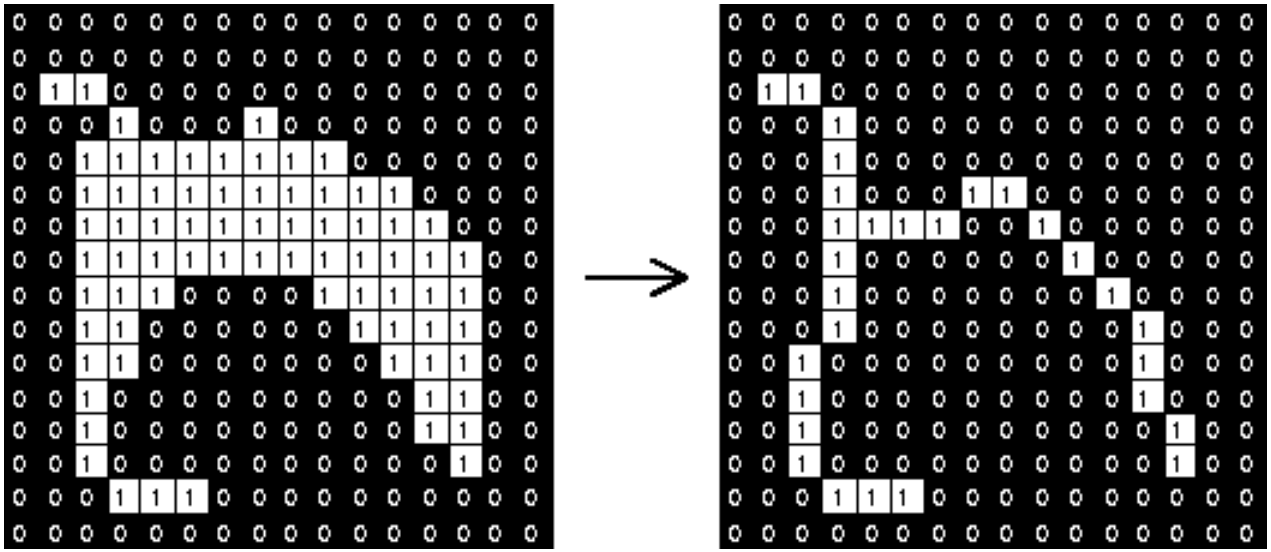
- Note that the value of the SE at the origin is 1 or *don't care*!

Example Thinning

0	0	0
	1	
1	1	1

	0	0
1	1	0
	1	

We use two Hit-and-miss Transforms



Basic
THICKENING

Thickening

- Used to grow selected regions of foreground pixels
- E.g. applications like approximation of *convex hull*

Definition Thickening

- Let ***K*** be a kernel and ***I*** be an image

$$\text{thicken}(I, K) = I + \text{HitAndMiss}(I, K)$$

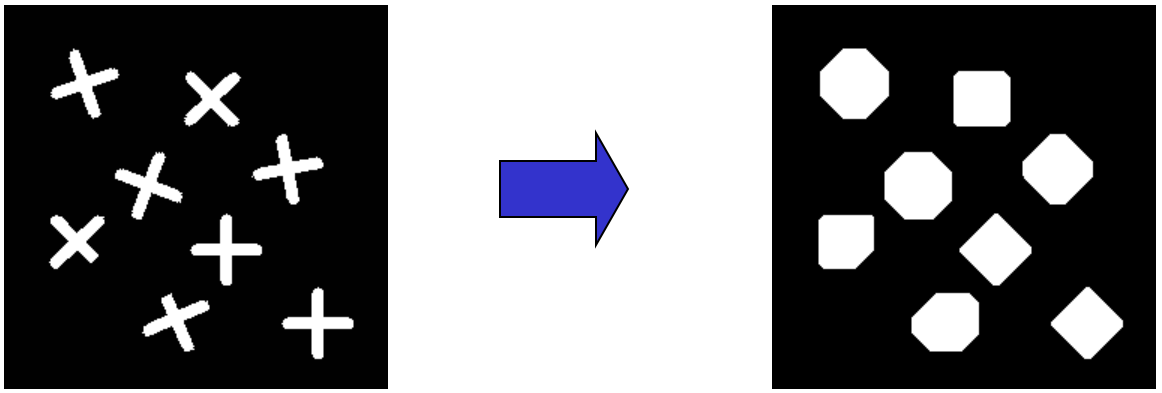
with 1+1=1

- If foreground and background match exactly the SE, then set the pixel at its origin to 1!
- Note that the value of the SE at the origin is 0 or *don't care*!

Example Thickening

1	1	
1	0	
1		0

	1	1
	0	1
0		1



PROBLEMS

Problems

1. Consider the images on slide 19! Why are the images getting darker under erosion? Explain!

2. Consider the images on slide 34! Why are the intensities becoming more uniform? Explain!

3. Compare Dilatation and Erosion! How are they related? Verify your answers with Matlab!

4. Apply Erosion and Dilatation for noise removal!

5. Consider slide 46: Remove the artefacts remaining with the horizontal lines.

Problems

6. Derive dilatation and erosion from the Hit-and-miss transformation

Sources Used

Volker Krüger & Rune Andersen
University of Aalborg, Esbjerg